you were outbound to Gargantua, to form a 24-solar-mass, nonspinning hole (with 6 solar masses of gas ejected during the implosion). Those two 24-solar-mass holes should now be circling around each other as a *black-hole binary,* and as they circle, they should emit ripples of tidal force (ripples of "spacetime curvature") called *gravitational waves.*[35] These gravitational waves should push back on the binary in much the same way as an outflying bullet pushes back on the gun that fires it, and this *gravitational-wave recoil* should drive the holes into a slow but inexorable inward spiral. With a slight adjustment of your starship's acceleration, you can time your arrival to coincide with the last stage of that inward spiral: Several days after you arrive, you will see the holes' nonspinning horizons whirl around and around each other, closer and closer, and faster and faster, until they coalesce to produce a single whirling, spinning, larger horizon.

Because the two parent holes do not spin, neither alone can serve as an efficient power source for your colony. However, the newborn, rapidly spinning hole will be ideal!

Home

After a 42-year voyage your starship finally decelerates into the Orion nebula, where DAWN predicted the two holes should be. There they are, right on the mark! By measuring the orbital motion of interstellar atoms as they fall into the holes, you verify that their horizons are not spinning and that each weighs 24 solar masses, just as DAWN predicted. Each horizon has a circumference of 440 kilometers; they are 30,000 kilometers apart; and they are orbiting around each other once each 13 seconds. Inserting these numbers into the general relativity formulas for gravitational-wave recoil, you conclude that the two holes should coalesce seven days from now. There is just time enough for your crew to prepare their telescopic cameras and record the details. By photographing the bright ring of focused starlight that encircles each hole's black disk, they can easily monitor the holes' motions.

You want to be near enough to see clearly, but far enough away to be safe from the holes' tidal forces. A good location, you decide, is a starship orbit ten times larger than the orbit in which the holes circle each other—an orbital diameter of 300,000 kilometers and orbital circumference of 940,000